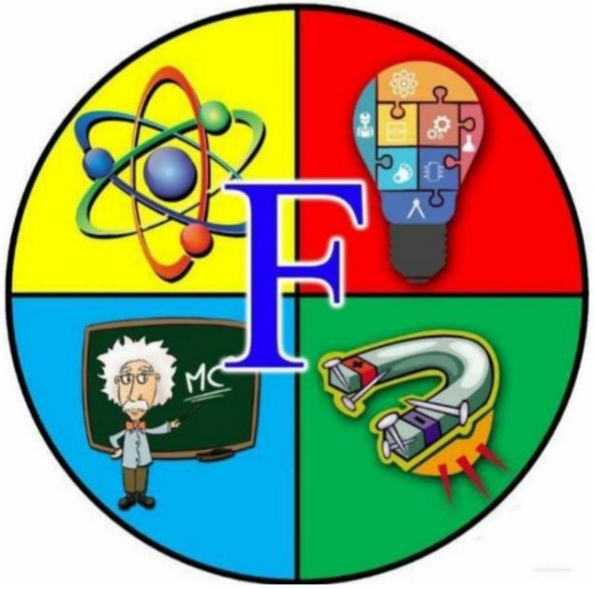
Fizika

9-sinf

Mavzu: **Masalalar yechish.**



Linzaning optik kuchi: $D = \frac{1}{F}; \quad F = \frac{1}{D}$

Fokus masofasi: $F = \frac{1}{(n-1)\cdot(\frac{1}{R_1}+\frac{1}{R_2})}$

$\frac{1}{F} = \frac{1}{d} + \frac{1}{f};$ $\quad -\frac{1}{F} = \frac{1}{d} - \frac{1}{f}$

Masala

Buyumni fokus masofasi 7,5 cm bo‘lgan yig‘uvchi linzadan 10 cm masofada joylashgan. Uning tasviri linzadan qanday masofada hosil bo‘ladi? Linzaning kattalashtirishi qanday?

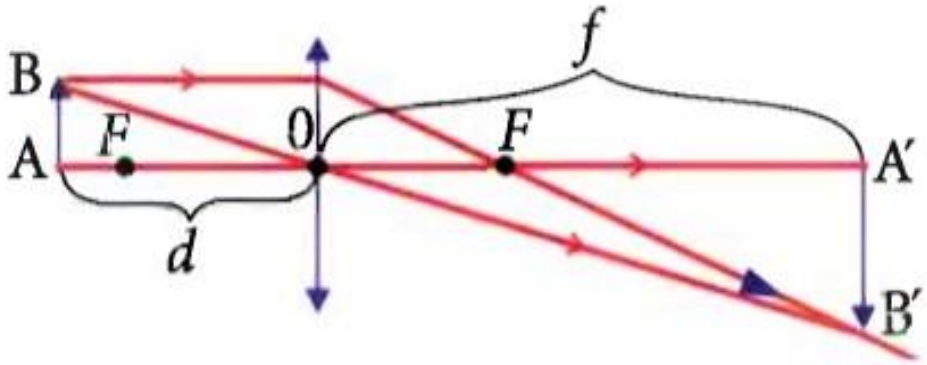
Berilgan:

$F = 7{,}5\ cm$

$d = 10\ cm$

Topish kerak: $f - ?$

Formula:



$$\frac{1}{F} = \frac{1}{d} + \frac{1}{f} \qquad f = \frac{dF}{d-F}$$

$$K = \frac{f}{d} = \frac{F}{d-F}$$

Yechish: $f = \frac{10 \cdot 7{,}5}{10 - 7{,}5}\ \text{cm} = 30\ \text{cm}$

$$K = \frac{30\ cm}{10\ cm} = 3$$

Javob: $\boldsymbol{f = 30\ cm;\ \ K = 3.}$

Masala

Linzadan 50 cm uzoqlikdagi jismning mavhum tasviri 2 marta kichiklashgan holda bo‘ldi. Linzaning optik kuchini toping.

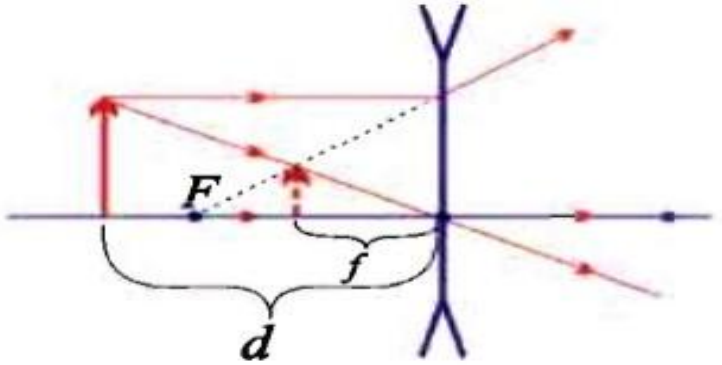
Berilgan:

$d = 50\ cm = 0{,}5\ m$

$K = 1/2$

Topish kerak: $D - ?$

Formula:



$$K = \frac{f}{d} \qquad f = d \cdot K$$

$$D = \frac{1}{d} - \frac{1}{f} = \frac{1}{d} - \frac{1}{d \cdot K} = \frac{K-1}{d \cdot K}$$

Yechish: $D = \frac{0,5-1}{0,5 \cdot 0,5}\ \text{dptr} = -2\ \text{dptr}$

Javob: $\boldsymbol{D = -2\ \text{dptr}}$

27-mashq 2-masala

Havoga nisbatan nur sindirish koʻrsatkichi 1,5 ga teng shishadan yasalgan sirtning egrilik radiuslari 20 cm va 25 cm boʻlgan ikkiyoqlama qavariq linzaning optik kuchini toping.

Berilgan:

$n = 1{,}5$

$R_1 = 20\ \text{cm} = 0{,}2\ \text{m}$

$R_2 = 25\ \text{cm} = 0{,}25\ \text{m}$

Topish kerak: $D - ?$

Formula:

$$D = (n-1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

Yechish: $$\text{D} = (1{,}5 - 1)\left(\frac{1}{0{,}2\ m} + \frac{1}{0{,}25\ m}\right) = 4{,}5\ \text{dptr}$$

Javob: $\boldsymbol{D = 4{,}5\ dptr}$

27-mashq 4-masala

O‘quvchi laboratoriya ishini bajara turib ekranda yonib turgan shamning aniq tasvirini hosil qildi.

Agar buyumdan linzagacha bo‘lgan masofa 15 cm, linzadan ekrangacha bo‘lgan masofa esa 60 cm bo‘lsa, linzaning fokus masofasi va optik kuchi qanday?

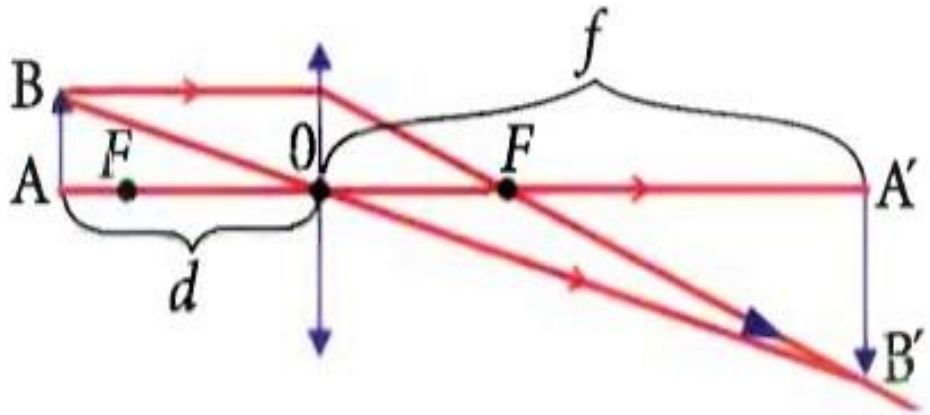
Berilgan:

$d == 15\ cm = 0{,}15\ m$

$f = 60\ cm = 0{,}6\ m$

Topish kerak: $F - ?\quad D - ?$

Formula:



$$\frac{1}{F} = \frac{1}{d} + \frac{1}{f} \qquad F = \frac{fd}{f+d} \qquad D = \frac{1}{F}$$

Yechish: $F = \dfrac{0{,}6\ m \cdot 0{,}15\ m}{0{,}6\ m + 0{,}15\ m} = 0{,}12\ \text{m} = 12\ \text{cm}$

$$D = \frac{1}{0{,}12\ m} = 8{,}33\ \text{dptr}$$

Javob: $\boldsymbol{F = 12\ cm;\quad D = 8{,}33\ dptr.}$

Masala

Buyumning mavhum tasviri linzadan 50 cm masofada hosil bo‘ldi.

Agar buyumdan linzagacha masofa 20 cm bo‘lsa, linzaning optik kuchini aniqlang.

Berilgan:

$f = 50\ cm = 0{,}5\ m$

$d = 20\ cm = 0{,}2\ m$

Topish kerak: $D - ?$

Formula:

$$D = \frac{1}{d} - \frac{1}{f} = \frac{f - d}{df}$$

Yechish: $D = \frac{0{,}5\ m - 0{,}2\ m}{0{,}2\ m \cdot 0{,}5\ m} = 3\ dptr$

Javob: $\boldsymbol{D = 3\ dptr}$

Mustaqil bajarish uchun topshiriqlar

149-bet **27-**mashq

3-, 5-, 7-, 8-masalalarni yechish.